1. Recall: an ideal $P \neq R$ in a commutative ring R is prime if $ab \in P$ implies $a \in P$ or $b \in P$.

   a) Prove that P is prime if and only if R/P is a domain.
   b) Use the first isomorphism theorem to show that the ideals (x) and $(2, x)$ in $\mathbb{Z}[x]$ are prime ideals.[1,2]
   c) Show that the ideal $(4, x)$ in $\mathbb{Z}[x]$ is not prime.
   d) Show that the ideal $(2, \sqrt{10})$ in $\mathbb{Z}[\sqrt{10}] = \{a + b\sqrt{10} \mid a, b \in \mathbb{Z}\} \subseteq \mathbb{R}$ is prime.
   e) Is the ideal (2) in $\mathbb{Z}[i]$ a prime ideal?

[1]Hint: For the first one, consider the homomorphism $\mathbb{Z}[x] \to \mathbb{Z}$ "evaluate at zero".
[2]Reminder: $(2, x)$ refers to the ideal generated by 2 and x.

(a) first we prove: if P is prime, then R/P is a domain. ①

Pf. Assume P is prime,

Let $a+P$, $b+P$ be two arbitrary elements in R/P, with $(a+P)(b+P) = 0_R+P$.

(note that 0_R+P is the additive identity in R/P)

So $ab+P = 0_R+P$

so $ab = ab - 0_R \in P$ by definition

Since P is prime, $a = 0_R$ or $b = 0_R$

So $a+P = 0_R+P$ or $b+P = 0_R+P$

i.e. $a+P = 0_{R/P}$ or $b+P = 0_{R/P}$.

So $(a+P)(b+P) = 0_{R/P}$ implies $a+P = 0_{R/P}$ or $b+P = 0_{R/P}$, so R/P is a domain.

Then we prove: if R/P is a domain, then P is prime ②

Pf. Assume R/P is a domain.

Let $a, b \in R$ be arbitrary with $ab \in P$

So $ab - 0_R \in P$

$ab + P = 0_R + P$

$(a+P)(b+P) = 0_R + P = 0_{R/P}$

Since R/P is a domain,

then $a+P = 0_{R/P}$ or $b+P = 0_{R/P}$

so $a - 0_R \in P$ or $b - 0_R \in P$

$a \in P$ or $b \in P$.

So $ab \in P \Rightarrow a \in P$ or $b \in P$

Therefore P is prime.

By ①②, we can conclude that P is prime iff R/P is a domain.

(b) Consider $\varphi: \mathbb{Z}[x] \longrightarrow \mathbb{Z}$

sending $f(x) \mapsto f(0)$.

Note that φ is a homomorphism. (already shown in worksheet)

and $\ker(\varphi) = \{kx \mid k \in \mathbb{Z}\} = (x)$.

Since $\forall z \in \mathbb{Z}$, $z \in \mathbb{Z}[x]$ so $\varphi(z) = z$,

φ is surjective.

So by first isomorphism Thm,

$\mathbb{Z}[x]/(x) \cong \mathbb{Z}$

Since $\mathbb{Z}$ is a domain, $\mathbb{Z}[x]/(x)$ is a domain since isomorphism preserves domain.

Then by (a), (x) is a prime ideal.

For $(2,x)$, Consider the function $\varphi: \mathbb{Z}[x] \mapsto \mathbb{Z}_2$

defined by $f(x) \mapsto [f(0)]_2$

We can show that this is a homomorphism: ①

Let $a = a_0 + a_1x + \dots + a_nx^n$

$b = b_0 + b_1x + \dots + b_mx^m \in \mathbb{Z}[x]$ be arbitrary.

So $\varphi(a+b) = [a_0+b_0]_2 = \varphi(a) + \varphi(b)$

$\varphi(ab) = [a_0b_0 + 0 + 0 + \dots + 0]_2 = [a_0b_0]_2 = \varphi(a)\varphi(b)$

and also $\varphi(0_{\mathbb{Z}[x]}) = 0_{\mathbb{Z}_2}$

And notice that φ is surjective:

$\varphi(0) = [0]_2$, $\varphi(1) = [1]_2$.

So by first isomorphism Thm,

$\mathbb{Z}[x]/\ker\varphi \cong \mathbb{Z}_2$

Since $\mathbb{Z}_2$ is a domain (since 2 is prime),

$\mathbb{Z}[x]/\ker\varphi$ is a domain

So by (a), $\ker\varphi$ is a prime ideal.

Since $\ker\varphi = \{kx + 2p \mid k, p \in \mathbb{Z}\}$

$= (2, x)$

$\Rightarrow (2,x)$ is a prime ideal

(c) Counterexample: consider $a = b = 2$

$2 \notin (4, x)$

$2 \cdot 2 = 4 \in (4, x)$

So $\exists\, a \notin P, b \notin P$ but $ab \in P$,

showing that P is not prime.

(d) Let $x = a + b\sqrt{10}$

$y = c + d\sqrt{10}$ $(a, b, c, d \in \mathbb{Z})$

be arbitrary elements of $\mathbb{Z}[\sqrt{10}]$

Assume $xy \in (2, \sqrt{10})$

so $xy = 2m + \sqrt{10}\,n$ for some integer $m, n \in \mathbb{Z}$

$\Rightarrow ac + (bc+ad)\sqrt{10} + 10bd = 2m + \sqrt{10}\,n$

$ac + 10bd = 2m$

$bc + ad = n$

Assume a, c are both odd for contradiction

So ac is odd

Since $10bd$ is even, $ac + 10bd$ is odd

contradicting with $ac + 10bd = 2m$

So at least one of a, c is even.

Without loss of generality, let a be even, so $a = 2k$ for some $k \in \mathbb{Z}$.

Therefore $x = 2k + b\sqrt{10} \in (2, \sqrt{10})$

So $xy \in (2, \sqrt{10})$ implies at least one of $x, y \in (2, \sqrt{10})$

So $(2, \sqrt{10})$ is a prime ideal on $\mathbb{Z}[\sqrt{10}]$.

(e) It is not a prime ideal.

Counterexample:

Consider $x = y = 1 + i$

So $xy = 1 + 2i - 1 = 2i \in (2)$

but $x, y \notin (2)$

2. We say that a proper ideal I in a ring R is **maximal** if whenever $I \subsetneq J$ for some ideal J, we have $J = R$. For the next problems, assume R is a commutative ring and I is an ideal of R.

   (a) Prove that if I is a maximal ideal and $a \notin I$ then $a + I$ is a unit in R/I.

   (b) Prove that I is a maximal ideal if and only if R/I is a field.[3]

   (c) Use the First Isomorphism Theorem to show that the non-principal ideal $(2, x)$ in $\mathbb{Z}[x]$ is a maximal ideal.[4]

   (d) Show that the ideal $(4, x)$ in $\mathbb{Z}[x]$ is not maximal.

   (e) Show that the ideal $(2, \sqrt{10})$ in $\mathbb{Z}[\sqrt{10}] := \{a + b\sqrt{10} \mid a, b \in \mathbb{Z}\} \subseteq \mathbb{R}$ is maximal.

   (f) Show that $I := \{a + bi \,:\, 3 \mid a \text{ and } 3 \mid b\}$ is a maximal ideal in $\mathbb{Z}[i]$.[5]

[3]Remark: Perhaps surprisingly, both directions of this "if and only if" are useful.

[4]Hint: Consider the homomorphism $f\colon \mathbb{Z}[x] \to \mathbb{Z}_2$ given by $f(x) \mapsto [f(0)]_2$

[5]Hint: If $r + si \notin I$, then $3 \nmid r$ or $3 \nmid s$. Show that 3 does not divide $r^2 + s^2 = (r + si)(r - si)$. Then show that and ideal containing $r + si$ and I also contains 1.

(a) Pf. We can construct a new ideal of R by $J = [I, a) = \{i + ak \mid i \in I, k \in R\}$

We can prove this is an ideal:

① Let $x = i_1 + ak_1$, $y = i_2 + ak_2$ be two arbitrary elements in J, so $x + y = (i_1 + i_2) + a(k_1 + k_2)$

Since $i_1 + i_2 \in I$, $x + y \in J$.

② Let $x = i + ak$ be an arbitrary element in J and r be an arbitrary element in R

so $x \cdot r = r \cdot x = ri + (kr)a$

Since $ri \in I$, $kr \in R$, $xr = rx \in J$

③ $0 \in J$

So J is an ideal of R

Note that $a \in J$, so since I is a maximal ideal, $J = R$.

So we can conclude: $\forall r \in R$, $r = i + ak$ for some $i \in I$ and $k \in R$

Then consider $1_R \in R$, $1_R = i + ak$ for some i.

Fix the i, so $ak = 1_R - i$

So $ak + I = 1_R - i + I$

Since $i \in I$, $-i + I = 0_R + I$

So $ak + I = 1_R + I$

$(a + I)(k + I) = 1_R + I$

Therefore $a + I$ is a unit in R/I.

(b) First we prove: if I is a maximal ideal, then R/I is a field. ①

This proof is almost finished by (a).

Since $R/I = \{a + I \mid a \in R\}$

$a + I = 0_{R/I}$ iff $a \in I$ and

for all $a \notin I$, $a + I$ is a unit by (a),

We have every non-zero element in R/I is a unit.

So R/I is a field. ① proved.

Then we prove: if R/I is a field, then I is a maximal ideal. ②

Assume R/I is a field.

Let J be an ideal of R such that $I \subsetneq J \subseteq R$

So $J \setminus I \neq \emptyset$, Let $a \in J \setminus I$ be arbitrary

Since R/I is a field,

there exists $b + I \in R/I$ s.t.

$(a + I)(b + I) = 1_R + I$.

so $ab - 1_R \in I \subsetneq J$

Since J is an ideal, $ab \in J$, So $1_R \in J$

Therefore $\forall r \in R$, $1_R \cdot r = r \in J$ by definition of ideal, so $R \subseteq J$

And since $R \subseteq J$, $J \subseteq R$ $\Rightarrow R = J$

So we have proved whenever $J \supsetneq I$ is an ideal, $J = R$

Hence I is a maximal ideal. ② proved.

By ①, ②, we can conclude: I is a maximal ideal iff R/I is a field.

(c) Consider the function $\varphi: \mathbb{Z}[x] \mapsto \mathbb{Z}_2$
defined by $f(x) \mapsto [f(0)]_2$

We have shown this is a homomorphism in 1(b),

Let $a = a_0 + a_1x + \dots + a_nx^n$
$b = b_0 + b_1x + \dots + b_mx^m \in \mathbb{Z}[x]$ be arbitrary.

So $\varphi(a+b) = [a_0+b_0]_2 = \varphi(a) + \varphi(b)$
$\varphi(ab) = [a_0b_0 + 0 + 0 + \dots + 0]_2 = [a_0b_0]_2 = \varphi(a)\varphi(b)$
and also $\varphi(0_{\mathbb{Z}[x]}) = 0_{\mathbb{Z}_2}$

And notice that φ is surjective:
$\varphi(0) = [0]_2$, $\varphi(1) = [1]_2$.

So by first isomorphism Thm,
$\mathbb{Z}[x]/\ker\varphi \cong \mathbb{Z}_2$

Since $\mathbb{Z}_2$ is a field ($[1][1] = [1]$),
$\mathbb{Z}[x]/\ker\varphi$ is a field (isomorphism preserves field)
So $\ker\varphi = (2, x)$ is a maximal ideal

(d) $(2,x) \neq \mathbb{Z}[x]$, is an ideal of $\mathbb{Z}[x]$
Notice that $(2,x) = \{2a + xb \mid a, b \in R\}$
$(4,x) = \{4a + xb \mid a, b \in R\}$
$= \{2(2a) + xb \mid a, b \in R\}$
$= \{2c + xb \mid b \in R, c = 2a, a \in R\}$

So $(4,x) \subsetneq (2,x)$
Therefore $(4,x)$ is not a maximal ideal in $\mathbb{Z}[x]$.

(e) Consider the quotient ring $\mathbb{Z}[\sqrt{10}]/(2, \sqrt{10})$

Let $a + b\sqrt{10} + I$ be an arbitrary element in it.
Since $b\sqrt{10} \in I$, $a + b\sqrt{10} + I = a + I$
Since $\forall k \in \mathbb{Z}$, $2k \in I$,
denote the remainder when a is divided by 2 as r, so $r = 1$ or 2
Then $a + I = r + I$
So $\mathbb{Z}[\sqrt{10}]/(2, \sqrt{10}) = \{1 + I, 0 + I\}$
which have only two elements.
This is a field since it is a commutative ring and the only nonzero element has a multiplicative inverse which is itself. $(1 + I)$

So $(2, \sqrt{10})$ is a maximal ideal.

(f) Let J be an ideal such that $I \subsetneq J \subseteq \mathbb{Z}[i]$
So there exist some $r + si \in \mathbb{Z}[i]$ such that $r + si \in J \setminus I$
i.e. either $3 \nmid r$ or $3 \nmid s$. or both

Since J is an ideal, $(r+si)(r-si) = r^2 + s^2 \in J$

Since either $3 \nmid r$ or $3 \nmid s$ or both,
$r \equiv a \mod 3$
$s \equiv b \mod 3$ $\Rightarrow$ $r^2 \equiv a^2 \mod 3$, $s^2 \equiv b^2 \mod 3$
where $a, b = 0$ or 1 or 2 $\Rightarrow a^2, b^2 = 0$ or 1.
and at least one of a, b is not 0.
so $r^2 + s^2 \equiv a^2 + b^2 \mod 3 \equiv 1 \mod 3$ or $2 \mod 3$.
so $3 \nmid r^2 + s^2$
So $\gcd(r^2 + s^2, 3) = 1$

$\Rightarrow$ by Bezóur, $x(r^2+s^2) + 3y = 1$
for some $x, y \in \mathbb{Z} \subseteq \mathbb{Z}[i]$

Since J is an ideal, $x(r^2+s^2) \in J$
and since $I \subsetneq J$, $3y \in J$, so
$x(r^2+s^2) + 3y \in J$
$\Rightarrow 1 \in J$

Then $\forall a \in \mathbb{Z}[i]$, $a \in J$, so $\mathbb{Z}[i] = J$
Therefore the only ideal J such that $I \subsetneq J$ is $J = \mathbb{Z}[i]$
So I is a maximal ideal.

3. Polynomial rings in *many* variables. Let $R_n = \mathbb{Q}[x_1, x_2, \dots, x_n]$ is a polynomial ring in variables $x_1, x_2, \dots, x_n$; that is it contains all polynomials in finite terms that involve these variables.

(a) Let $f_1, f_2, \dots, f_k$ be polynomials in R_n. Prove that the set

$$\langle f_1, f_2, \dots f_k \rangle := \{g_1f_1 + g_2f_2 + \dots + g_kf_k \mid g_1, g_2, \dots, g_k \in R_n\}$$

is an ideal of R_n.

(b) Consider the ring homomorphism

$$\varphi : R_4 \to \mathbb{Q}[t_1, t_2], \varphi(x_1) = t_1^3,\ \varphi(x_2) = t_1^2t_2,\ \varphi(x_3) = t_1t_2^2,\ \varphi(x_4) = t_2^3. \quad (1)$$

(c) Explain why the above description fully determines $\varphi(f)$ for each polynomial $f \in R_4$.

(d) It is given to you that $\ker(\varphi) = \langle f_1, f_2, f_3 \rangle$ for some polynomials $f_1, f_2, f_3 \in R_4$. Find f_1, f_2, f_3. *Hint: part (e) may help.*

(e) Let h_1, h_2, h_3 be the 2 by 2 minors of the matrix $M = \begin{bmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \end{bmatrix}$. Consider the ideal $I = \langle h_1, h_2, h_3 \rangle$. Show that I does not change if one applies elementary row operations to the matrix M.

(f) Take the ideal $J = \langle x_1x_4 - x_2x_3 \rangle$ in R_4. Express J as kernel of some ring homomorphism. You know such a homomorphism exists by WSH 10. You do not need to prove that the proposed homomorphism has J as its kernel. Hopefully you will be able to prove this by the end of the semester.

(g) Prove that the ideal J is not a maximal ideal.

(a) ① Select arbitrary $x = g_1f_1 + g_2f_2 + \dots + g_kf_k$,

$y = h_1f_1 + h_2f_2 + \dots + h_kf_k \in \langle f_1, f_2, \dots, f_k\rangle$

where $g_1,\dots,g_k,h_1,\dots,h_k \in R_n$

$\Rightarrow x+y = (g_1+h_1)f_1 + (g_2+h_2)f_2 + \dots + (g_k+h_k)f_k$

since $g_1+h_1,\dots,g_k+h_k \in R_n$, $x+y \in \langle f_1, f_2, \dots, f_k\rangle$

② Select arbitrary $x = g_1f_1 + g_2f_2 + \dots + g_kf_k \in \langle f_1, f_2, \dots, f_k\rangle$

and $h \in R_n$

$xh = (hg_1)f_1 + (hg_2)f_2 + \dots + (hg_k)f_k \in \langle f_1, f_2, \dots, f_k\rangle$

③ $0 = 0f_1 + 0f_2 + \dots + 0f_k \in \langle f_1, f_2, \dots, f_k\rangle$

By ①②③, $\langle f_1, f_2, \dots, f_k\rangle$ is an ideal in R_n.

(c) $\forall \underbrace{c}_{\text{constant}} \in R_4$, $\varphi(c) = \varphi(\overbrace{1_{R_4} + 1_{R_4} + \dots + 1_{R_4}}^{c \text{ times}})$

$= c\cdot\varphi(1_{R_4}) = c\cdot\varphi(1_{\mathbb{Q}})$

$= c$

For arbitrary element $f \in R_4$

$f = a_0 + a_1x_1 + a_2x_1^2 + \dots + a_ix_1^i$
$+ b_0 + b_1x_2 + b_2x_2^2 + \dots + b_jx_2^j$
$+ c_0 + c_1x_3 + \dots + c_mx_3^m$
$+ d_0 + d_1x_4 + \dots + d_nx_4^n$ for some

So $\varphi(f) = (a_0+b_0+c_0+d_0) + a_1\varphi(x_1) + \dots + a_i\varphi^i(x_1)$
$+ \dots + b_j\varphi^j(x_2) + \dots + c_m\varphi^m(x_3) + \dots + d_n\varphi^n(x_4)$

since homomorphism preserves addition and multiplication,

Each term is either constant in $\mathbb{Q}[t_1,t_2]$ or some constant multiplied by some multiples of $\varphi(x_1)$ or $\varphi(x_2)$ or $\varphi(x_3)$ or $\varphi(x_4)$

Hence $\varphi(f)$ is fully determined for each $f \in R_4$

(d) Consider $f_1 = x_1x_3 - x_2^2$

$f_2 = x_2x_4 - x_3^2$ $\quad$ ($\Rightarrow \varphi(f_1) = \varphi(f_2)$

$f_3 = x_1x_4 - x_2x_3$ $\quad$ $= \varphi(f_3) = \varphi(0) = 0$)

take arbitrary element $a \in \langle f_1, f_2, f_3\rangle$,

$a = g_1f_1 + g_2f_2 + g_3f_3$ for some $g_1, g_2, g_3 \in R_4$

So $\varphi(a) = \varphi(g_1f_1) + \varphi(g_2f_2) + \varphi(g_3f_3)$

$= \varphi(g_1)\cdot 0 + \varphi(g_2)\cdot 0 + \varphi(g_3)\cdot 0 = 0.$

(e) $h_1 = \begin{vmatrix} x_1 & x_2 \\ x_2 & x_3 \end{vmatrix} = x_1x_3 - x_2^2$

$h_2 = \begin{vmatrix} x_2 & x_3 \\ x_3 & x_4 \end{vmatrix} = x_2x_4 - x_3^2$

$h_3 = \begin{vmatrix} x_1 & x_3 \\ x_2 & x_4 \end{vmatrix} = x_1x_4 - x_2x_3$

$I = \langle h_1, h_2, h_3\rangle = \{g_1(x_1x_3 - x_2^2) + g_2(x_2x_4 - x_3^2) + g_3(x_1x_4 - x_2x_3) \mid g_1, g_2, g_3 \in R_4\}$

① Swapping the two rows does not change I

By swapping the rows we get $h_1' = x_2^2 - x_1x_3$

$h_2' = x_3^2 - x_2x_4$, $h_3' = x_2x_3 - x_1x_4$

So $I' = \{-g_1(x_1x_3 - x_2^2) - g_2(x_2x_4 - x_3^2) - g_3(x_1x_4 - x_2x_3) \mid g_1, g_2, g_3 \in R_4\}$

Since $g_1, g_2, g_3 \in I \Leftrightarrow -g_1, -g_2, -g_3 \in I$,

$I' = I$

② Multiplying a row by a nonzero constant does not change I.

WLOG assume we multiply row one by $a \in \mathbb{Q}$ ($a \neq 0$)

so $h_1' = a(x_1x_3 - x_2^2)$

$h_2' = a(x_2x_4 - x_3^2)$

$h_3' = a(x_1x_4 - x_2x_3)$

so $I' = \{a[g_1(x_1x_3 - x_2^2) + g_2(x_2x_4 - x_3^2) + g_3(x_1x_4 - x_2x_3)] \mid g_1, g_2, g_3 \in R_4\}$

Since $\mathbb{Q}$ is a field, $a \neq 0 \Rightarrow$ ($\frac{1}{a} \in \mathbb{Q}$ iff $a \in \mathbb{Q}$)

So $\frac{1}{a} \in R_4$ iff $a \in R_4$

So $a[g_1(x_1x_3 - x_2^2) + g_2(x_2x_4 - x_3^2) + g_3(x_1x_4 - x_2x_3)] \in I$

iff $g_1(x_1x_3 - x_2^2) + g_2(x_2x_4 - x_3^2) + g_3(x_1x_4 - x_2x_3) \in I$

by multiplying a and $\frac{1}{a}$ respectively.

So $I' = I$.

③ Adding some nonzero multiple of a row to another does not change I

WLOG assume we add a multiple of the second row to the first row.

$\to \begin{bmatrix} x_1 + bx_2 & x_2 + bx_3 & x_3 + bx_4 \\ x_2 & x_3 & x_4 \end{bmatrix}$, where $b \neq 0 \in \mathbb{Q}$

Then $h_1' = x_1x_3 - x_2^2 = h_1$

$h_2' = x_2x_4 - x_3^2 = h_2$

$h_3' = x_2x_3 - x_1x_4 = h_3$

So $I = \langle h_1, h_2, h_3 \rangle = \langle h_1', h_2', h_3' \rangle = I'$

By ①②③, we can conclude that I does not change if one applies elementary row operations to M.

(f) Consider

$\varphi : R_4 \longrightarrow \mathbb{Q}[x_2, x_3]$ defined by

① $\varphi(x_1) = x_2$, $\varphi(x_2) = x_2$, $\varphi(x_3) = x_3$, $\varphi(x_4) = x_3$

For the same reason as in (c), φ is determined by ①.

Then $J = \langle x_1x_4 - x_2x_3 \rangle = \{ g(x_1x_4 - x_2x_3) \mid g \in R_4 \}$ is $\ker\varphi$. $(x_1x_4 - x_2x_3 = 0)$

This homomorphism is well-defined, which is easy to see because ① $\varphi(1) = 1$, ② φ preserves addition in R_4 and ③ φ preserves multiplication in R_4, seen from the polynomial addition and multiplication operation.

(g) Consider $K = \langle x_4, x_2 \rangle$ as an ideal of R_4

so $x_1x_4 - x_2x_3 \in K$

Select arbitrary $g \in R_4$, by property of ideal, $g(x_1x_4 - x_2x_3) \in K$

So $\{ g(x_1x_4 - x_2x_3) \mid g \in R_4 \} = J \subseteq K$

And consider $x_4 + x_2 \in K$ but $x_4 + x_2 \notin J$,

so $x_4 + x_2 \in K \setminus J \Rightarrow J \subsetneq K$.

And consider $x_1 + x_3 \in R_4$ but $x_1 + x_3 \notin K$, $K \subsetneq R_4$

So $J \subsetneq K \subsetneq R_4$

By definition, J is not a maximal ideal.